

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i = \sqrt{-1}$. Then, the system of linear equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has :

- (1) A unique solution
- (2) Infinitely many solutions
- (3) No solution
- (4) Exactly two solutions

Q2: 16 March (Shift 1) - Numerical

The total number of 3×3 matrices A having entries from the set $(0,1,2,3)$ such that the sum of all the diagonal entries of AA^T is 9, is equal to _____

Q3: 16 March (Shift 1) - Numerical

Let

$$P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix} \text{ and } A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega + 1 \end{bmatrix}$$

where $\omega = \frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____

Q4: 16 March (Shift 2) - Numerical

Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that $A = XB$,

where $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and $k \in \mathbb{R}$.

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If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$,

then the value of k is _____

Q5: 17 March (Shift 1) - Single Correct

If $A = \begin{pmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{pmatrix}$ and $\det(A^2 - \frac{1}{2}I) = 0$, then a possible value of α is

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{6}$

Q6: 17 March (Shift 1) - Numerical

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to _____

Q7: 17 March (Shift 2) - Numerical

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that

$AB = B$ and $a + d = 2021$, then the value of $ad - bc$ is equal to _____.

Q8: 18 March (Shift 1) - Single Correct

Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$. If $\text{Tr}(A)$ denotes the sum of all diagonal elements of the matrix A , then $\text{Tr}(A) - \text{Tr}(B)$ has value equal to

(1) 1

(2) 2

(3) 0

(4) 3

Questions with Answer Keys

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Q9: 18 March (Shift 2) - Numerical

Let I be an identity matrix of order 2×2 and

$P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of $n \in \mathbb{N}$ for which $P^n = 5I - 8P$ is equal to ____

Answer Key**Q1 (3)****Q2 (766)****Q3 (36)****Q4 (1)****Q5 (3)****Q6 (16)****Q7 (2020)****Q8 (2)****Q9 (6)**

Q1 (3)

$$A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 2 \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^4 = 2^2 \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 8 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^8 = 64 \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 128 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow 128 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow 128 \begin{bmatrix} x - y \\ -x + y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow x - y = \frac{1}{16}$$

$$\& -x + y = \frac{1}{2}$$

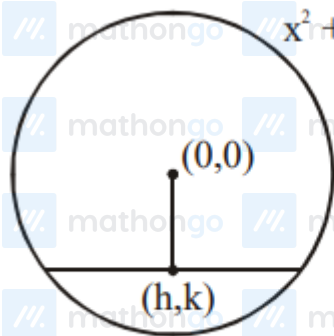
Q2 (766)

$$\text{Let } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

diagonal elements of AA^T , $a^2 + b^2 + c^2$, $d^2 + e^2 + f^2$, $g^2 + h^2 + i^2$

$$\text{Sum} = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 9$$

$$a, b, c, d, e, f, g, h, i \in \{0, 1, 2, 3\}$$



Equation of chord

$$\text{Total no. of ways} = 1 + 9 + 8 \times 63 + 63 \times 4 = 766$$

Q3 (36)

$$\begin{aligned}\text{Let } M &= (P^{-1}AP - I)^2 \\ &= (P^{-1}AP)^2 - 2P^{-1}AP + I \\ &= P^{-1}A^2P - 2P^{-1}AP + I\end{aligned}$$

$$\begin{aligned}PM &= A^2P - 2AP + P \\ &= (A^2 - 2A \cdot I + I^2)P \\ \Rightarrow \text{Det}(PM) &= \text{Det}((A - I)^2 \times P) \\ \Rightarrow \text{Det}P \cdot \text{Det}M &= \text{Det}(A - I)^2 \times \text{Det}(P)\end{aligned}$$

$$\begin{aligned}\Rightarrow \text{Det } M &= (\text{Det}(A - I))^2 \\ \text{Now } A - I &= \begin{bmatrix} 1 & 7 & w^2 \\ -1 & -w-1 & 1 \\ 0 & -w & -w \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\text{Det}(A - I) &= (w^2 + w + w) + 7(-w) + w^3 = -6w \\ \text{Det}((A - I))^2 &= 36w^2 \\ \Rightarrow \alpha &= 36\end{aligned}$$

Q4 (1)

$$\begin{aligned}A &= XB \\ \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} &= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ \begin{bmatrix} \sqrt{3}a_1 \\ \sqrt{3}a_2 \end{bmatrix} &= \begin{bmatrix} b_1 - b_2 \\ b_1 + kb_2 \end{bmatrix} \\ b_1 - b_2 &= \sqrt{3}a_1 \dots (1)\end{aligned}$$

$$b_1 + kb_2 = \sqrt{3}a_2 \dots (2)$$

$$\begin{aligned}\text{Given, } a_1^2 + a_2^2 &= \frac{2}{3}(b_1^2 + b_2^2) \\ (1)^2 + (2)^2\end{aligned}$$

$$(b_1 + b_2)^2 + (b_1 + kb_2)^2 = 3(a_1^2 + a_2^2)$$

$$a_1^2 + a_2^2 = \frac{2}{3}b_1^2 + \frac{(1+k^2)}{3}b_2^2 + \frac{2}{3}b_1b_2(k-1)$$

$$\text{Given, } a_1^2 + a_2^2 = \frac{2}{3}b_1^2 + \frac{2}{3}b_2^2$$

On comparing we get

Hints and Solutions

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$$\frac{k^2+1}{3} = \frac{2}{3} \Rightarrow k^2 + 1 = 2$$

$$\Rightarrow k = \pm 1$$

$$\& \frac{2}{3}(k-1) = 0 \Rightarrow k = 1$$

From both we get $k = 1$

Q5 (3)

$$A^2 = \sin^2 \alpha I$$

$$\text{So, } |A^2 - \frac{I}{2}| = \left(\sin^2 \alpha - \frac{1}{2}\right)^2 = 0$$

$$\Rightarrow \sin \alpha = \pm \frac{1}{\sqrt{2}}$$

Q6 (16)

$$2 A \operatorname{adj}(2 A) = 12 A I$$

$$\Rightarrow A \operatorname{adj}(2 A) = -4 I \quad \dots (i)$$

$$\text{Now, } E = |A^4| + |A^{10} - (\operatorname{adj}(2 A))^{10}|$$

$$= (-2)^4 + \frac{|A^{20} - A^{10}(\operatorname{adj} 2 A)^{10}|}{|A|^{10}}$$

$$= 16 + \frac{|A^{20} - (A \operatorname{adj}(2 A))^{10}|}{|A|^{10}}$$

$$= 16 + \frac{|A^{20} - 2^{10} I|}{2^{10}} \quad (\text{from } (i))$$

Now, characteristic roots of A are 2 and -1. So, characteristic roots of A^{20} are 2^{10} and 1.

$$\text{Hence, } (A^{20} - 2^{10} I)(A^{20} - I) = 0$$

$$\Rightarrow |A^{20} - 2^{10} I| = 0 \quad (\text{as } A^{20} \neq I)$$

$$\Rightarrow E = 16 \text{ Ans.}$$

Q7 (2020)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$AB = B$$

$$\Rightarrow (A - I)B = O$$

$$\Rightarrow |A - I| = 0, \text{ since } B \neq O$$

Hints and Solutions

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$$\begin{vmatrix} (a-1)b & c \\ c & (d-1)a \end{vmatrix} = 0$$

$$ad - bc = 2020$$

Q8 (2)

$$A + 2B = \begin{pmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{pmatrix} \dots (1)$$

$$2A - B = \begin{pmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow 4A - 2B = \begin{pmatrix} 4 & -2 & 10 \\ 4 & -2 & 12 \\ 0 & 2 & 4 \end{pmatrix} \dots (2)$$

$$(1) + (2) \Rightarrow 5A = \begin{pmatrix} 5 & 0 & 10 \\ 10 & -5 & 15 \\ -5 & 5 & 5 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ -1 & 1 & 1 \end{pmatrix} \text{ and } 2A = \begin{pmatrix} 2 & 0 & 4 \\ 4 & -2 & 6 \\ -2 & 2 & 2 \end{pmatrix}$$

$$\therefore B = \begin{pmatrix} 2 & 0 & 4 \\ 4 & -2 & 6 \\ -2 & 2 & 2 \end{pmatrix} - \begin{pmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 1 & -1 \\ 2 & -1 & 0 \\ -2 & 1 & 0 \end{pmatrix}$$

$$\text{tr}(A) = 1 - 1 + 1 = 1$$

$$\text{tr}(B) = -1$$

$$\text{tr}(A) = 1 \text{ and } \text{tr}(B) = -1$$

$$\therefore \text{tr}(A) - \text{tr}(B) = 2$$

Q9 (6)

$$P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$$

$$5I - 8P = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 16 & -8 \\ 40 & -24 \end{bmatrix} = \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} -1 & 1 \\ -5 & 4 \end{bmatrix}$$

Hints and Solutions

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$$P^3 = \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix} \Rightarrow P^6 = \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix} = P^n$$
$$\Rightarrow n = 6$$